

CoDec

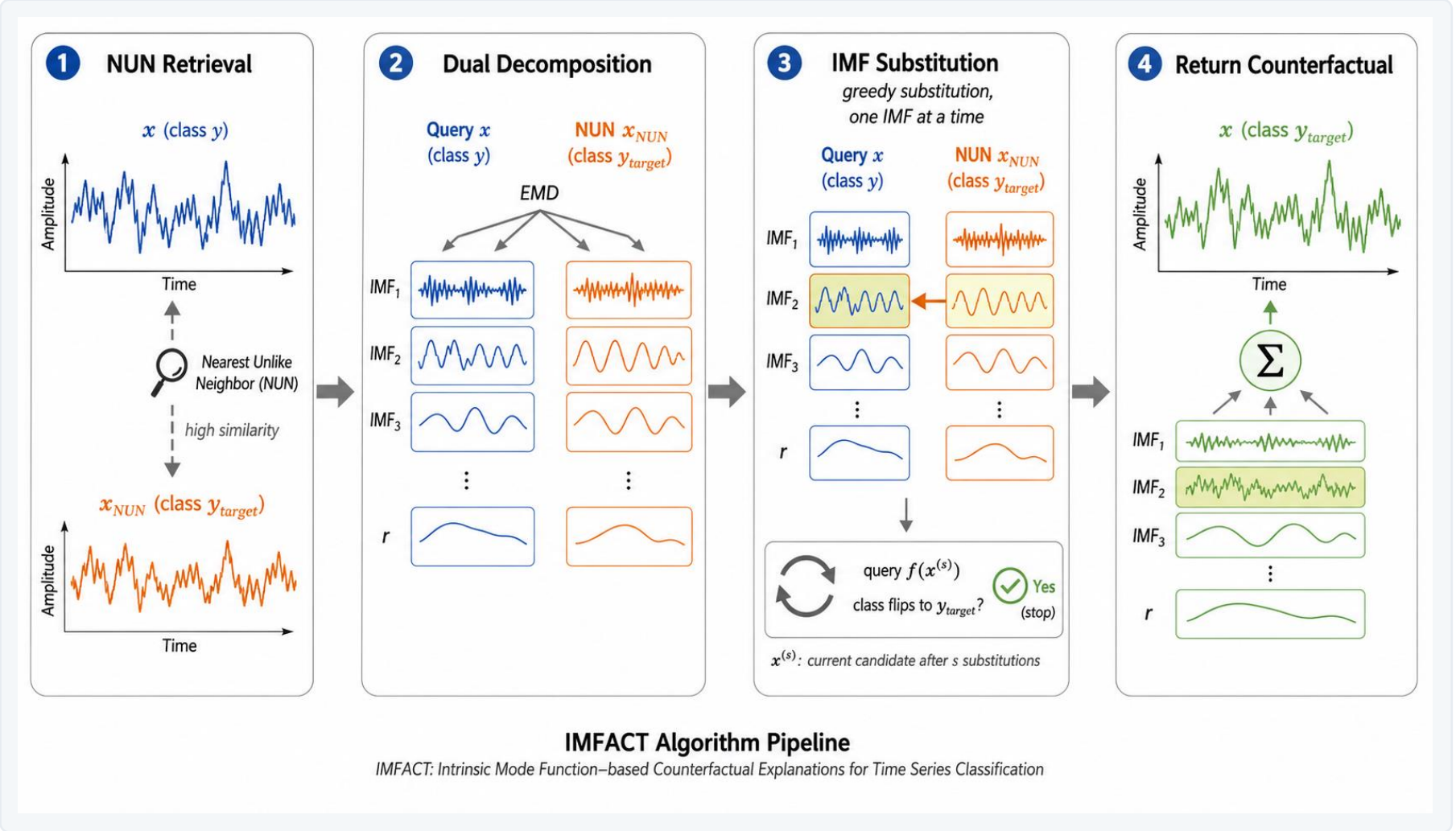
Counterfactual Decomposition

A modular framework for decomposition-based counterfactual explanations of time series classifiers — extending IMFACT

encode → intervene → decode

Starting Point: IMFACT

The single-decomposition, IMF-specific pipeline that CoDec generalizes



Motivation

Why generalize IMFACT into a framework

The problem

Time series classifiers are black boxes. Counterfactual explanations give actionable interpretability — the minimal change that would flip the prediction.

The gap

Existing methods either perturb raw time points, risking implausible series, or commit to one fixed decomposition that doesn't generalize across datasets.

What IMFACT showed

Decomposition-based counterfactuals work — but reviewers flagged it as tied to a single method, not a validated general approach.

The move

Generalize IMFACT into CoDec: a modular framework where decomposition, matching, and perturbation are all swappable and empirically validated. Inform decomposition based on time series characteristics

Why not “Spectral”?

The umbrella term has to survive every decomposition we plug in

“Spectral” has two narrow senses:

- Frequency-domain — Fourier, wavelets, DCT
-
- Eigen-decomposition — PCA, SVD, SSA
-
-

EMD / IMF is adaptive & data-driven — it only becomes “spectral” after a subsequent Hilbert transform. It doesn’t cleanly belong to either box.

Other candidates aren’t spectral at all: STL, changepoint segmentation, SAX, state-space models, quantile transforms.

Two distinct categories — EMD sits outside both



The CoDec Pipeline

Eight modular stages, grouped into three conceptual acts

ENCODE

1

Reference selection

k-NUN candidates from target class

2

Decomposition

EMD, wavelets, PCA, STL, ...

3

Matching

align components (Hungarian?)

INTERVENE

4

Component selection

which components to change

5

Perturbation

replace / interpolate / multi-ref

DECODE

6

Reconstruction

recompose the candidate series

7

Classifier query

check validity, loop if needed

8

Best-CF selection

proximity, plausibility, sparsity

The CoDec Approach

IMFACT's four stages, generalized: any reference, any decomposition, any match

1

Reference Selection

Query x (class y)





select reference — NUN, composite, or barycenter

Reference x_{ref} (class y_{target})



IMFACT used: NUN retrieval only

2

Component Decomposition

Query

Reference

decompose (D) — any method

c_1



c_2



c_3



$\vdots$

r



IMFACT used: EMD only

3

Component Perturbation

Query

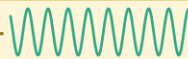
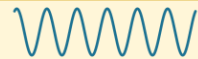
Reference

match (Hungarian)

c_1



c_2



c_3



$\vdots$

r



$\hookrightarrow$ retry loop: query $f(x')$

class flips to target? $\rightarrow \checkmark$ stop

else: reselect components or try next reference

IMFACT used: greedy, index-matched

4

Return Counterfactual

x' (class y_{target})





recomposes every component — only c_2 changed; c_1 and c_3 pass through untouched

IMFACT used: direct IMF summation

Optional: Beyond a Single NUN

Decoupling proximity from validity when choosing references

Composite reference

FAVORITE

Per component, borrow from whichever class instance matches best. No single donor series — a stitched, arbitrary-per-slot NUN.

Barycenter reference

Average / DTW-barycenter components across k-NUN into a synthetic prototype.

Typicality-anchored

Pick by class density (LOF, isolation forest), not distance to x.

Generative reference

Sample a class-consistent component directly — higher risk, later-stage idea.

Module Spotlight: Matching

Solving cross-series component alignment with Hungarian assignment

The problem

Decompositions can produce different numbers of components, and component indices aren't reliably aligned across series — a key reviewer criticism.

The fix

Build a cost matrix $C[i,j]$ = unsuitability of matching query component i to reference component j . Hungarian assignment finds the minimum-cost one-to-one matching.

Cost can be built from dominant frequency, spectral similarity, energy, or simple index alignment.

	r_1	r_2	r_3	r_4
c_1			✓	
c_2	✓			
c_3				✓

minimum-cost assignment

Choosing a Decomposition

A heuristic for matching the transform to what actually characterizes the dataset

Series characteristic	Best-fit decomposition	Why it fits
Multi-scale, non-stationary oscillations	EMD / Wavelets (EWT)	Adaptive, data-driven band separation — no fixed basis assumed
Strong trend + seasonal structure	STL / Fourier (STFT)	Explicit trend / seasonal / residual split
Correlated multivariate signals	PCA / SSA	Captures the dominant shared modes of joint variance
(Optional) Discriminative local shape (e.g. ECG, gesture)	Shapelets	Preserves the exact sub-sequence shape driving the decision
(Optional) Abrupt regime shifts or level changes	Changepoint segmentation	Localizes shift points directly; smooth bases blur them
(Optional) Heavy noise, few clean repeating patterns	Quantile transforms / state-space	Robust to noise; models uncertainty, not just a point curve

First characterize the series — dominant frequency, stationarity, morphology, noise level — then match to the row above. Every choice stays swappable, per CoDec’s modular design.

Experimental Plan

Tightening baselines, datasets, and metrics ahead of submission

Baselines	Add recent time-series CF algorithms; lean toward a small, well-justified subset over an exhaustive table.
Datasets	128 UCR + 30 UEA Full-archive evaluation — every univariate UCR and multivariate UEA dataset, not a hand-picked subset. Avoids cherry-picking criticism (later on). <i>Characteristic spotlights within the archive, tied to the decomposition heuristic:</i> CBF <i>shape-driven</i> ECG5000 <i>morphology-critical</i> FordA <i>frequency-heavy, noisy</i> Yoga <i>subtle amplitude / shape</i> TwoPatterns <i>synthetic, controlled</i>
Metrics	Report sparsity by components changed, not raw time points — perturbing a component can shift the whole series.
Classifier scope	Keep it black-box — no gradients. Sell as an advantage (classifier-agnostic); note classifier-aware perturbation only as future work.

Next Steps

Udo to lead coordination — next meeting sometime in September

1

Name & scope

Lock in CoDec as the framework name; verify no collisions before final commitment.

2

Next meeting

Sometime in September — these are the steps until then:

- ☐ Prototype the composite reference-selection module (favored arbitrary-NUN generalization)
- ☐ Finalize the small, well-justified baseline CF algorithm subset
- ☐ Stand up the full UCR + UEA archive evaluation pipeline
- ☐ Draft the Hungarian-matching implementation for the matching module

encode → intervene → decode